LAW AND SECURITY IN OUTER SPACE: IMPLICATIONS FOR PRIVATE ENTERPRISE +

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Coverage of this topic will be from the point of view of 1) law for national security, 2) applicable treaties, particularly the 1967 Outer Space Principles Treaty[1] and the 1979 Moon Treaty,[2] as well as 3) Private Enterprise and Unispace '82[3] and 4) Financial Incentives.[4]

Outer Space Peace is essential to private enterprise. It is always wise to look historically, even in the relatively new field of international law of outer space. In July, 1970, the American Bar Association published my article "Arms Control is not Disarmament."[5] In that article, I emphasized the importance of a nuclear balance of power in outer space and that outer space holds the keys to world peace. Today, we see again these two basic concepts right up front.

In 1969, at the 16th Convention in Caracas, Venezuela, of the Inter-American Bar Associaton, my published article "Space Liability and World Peace,"[6] also emphasized these two problems. Also in my article, "Outer Space for Peaceful Purposes,"[7] these problems were discussed in relation to the question of the semantic problem of the meaning of "military" peaceful, and non-aggressive, in five languages. We should pause now to again examine the question of outer space nuclear balance of power. In 1967, the Outer Space Principles Treaty clearly prohibits the orbiting of nuclear weapons of mass destruction. It does not prohibit the orbiting of conventional weapons and, in fact, Russia in some fifteen ASAT[8] tests in the last ten years has used a

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+ The views expressed in this article are those of the author and not necessarily those of the International Academy of Astronautics.

[1]Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, Including the Moon and Other Celestial Bodies January 27, 1967 [1967], 18 U.S.T. 2410, T.I.A.S. 6347, 610 U.N.T.S. 205 (effective Oct. 10, 1967).

[2]Draft Agreement Governing the Moon and Other Celestial Bodies U.N. GAOR, 34th Sess., Supp. H. No. 20 (Doc. A/34/20).

[3]A/CONF/10//NP/53 (March 23, 1982).

[4]20 A.B.A.J. 40 (May 1982).

[5]Finch, *Arms Control is not Disarmament*, 4 INT'L LAWYER (July 1970).

[6]"Space Liability and World Peace", speech made by author before A.B.A. Annual Convention, Aug. 4, 1978.

[7]54 A.B.A.J. 365 (1968).

[8]The term "anti-satellite" (ASAT) is generally used to describe any device that can be used to destroy the operational capability of satellites in earth orbit. These devices can be based on the ground, in airplanes, or in space.

conventional explosion in proximity to another of their satellites to test the operational ASAT capability. Happily, the approximately ninety signatories to the 1967 Outer Space Principles Treaty have observed the prohibition against the orbiting of nuclear weapons of mass destruction or of any weapon specifically for mass destruction and it is hoped that this will continue indefinitely for world peace. We must never forget that in outer space simple debris can become a very serious high velocity weapon of destruction to other satellites in outer space.

Further, with regards to the nuclear balance of power in outer space, many scientists and lawyers believe that Einstein is, for pragmatic purposes, a real keeper of world peace. As of April 7, 1982, without a doubt, manned space stations and satellite reconnaissance objects of the United States and Russia are the vigilant eyes of the two major space powers to be sure that none is launching nuclear weapons of mass destruction against any other nation. On April 7, 1982, the United States had 425 near earth satellites, plus 30 space probes. The United States also had 2,185 pieces of space debris or junk and 44 pieces of space probe debris. Total United States objects in outer space were 2,684. In context, Russia had 642 near earth satellites and 25 outer space probes. Russia had 1,127 pieces of space debris or junk and 111 space probe debris. Russia's total was 1,805. Grand total for all 18 nations currently in outer space were 1,173 near earth satellites, 57 space probes, 3,365 pieces of space junk or debris and 56 pieces of space probe debris. Thus, there are 4,651 objects altogether in outer space, as of April 7, 1982.[9] Further examining the nuclear outer space balance of power, there is currently under discussion in the Committee on Peaceful Uses of Outer Space in the United Nations, an international satellite monitoring agency to use satellite reconnaissance data to verify treaty compliance for world peace.

The importance of keeping outer space peaceful, was also recently re-emphasized by Ambassador Peter Jankowitsch, current Chairman of COPUOS, on the day of the successful third United States Shuttle Launch. He stated "The world has maintained a nuclear free zone in outer space, free of national sovereignty and should remain so." Other than the ASAT developments of the United States and Russia, and the danger arising from outer space debris which could constitute a "weapon", there is presently the destabilizing factor of the continuing persistent demands of the lesser developed countries to gain benefits without cost to them from outer space. To meet this, both the United States and Russia in Unispace '82 mounted considerable exhibits, together with lectures and films there to educate the lesser developed countries.

It would serve no useful purpose to elaborate on the second major principle of my remarks today, mainly that outer space is the keeper of world peace. The reconnaissance satellite proposals already discussed speak loud and clear to that principle, as does existing outer space technology for medicine, food and minerals, etc.

With regard to arms control in outer space, little progress has been made in the ASAT limitation meetings by the United States and Russia, which have been held from time to time in recent years. It is my humble prediction that until the technology of ABM Laser Beam Defense and Particle Beam Defense have exhausted considerably more of their scientific and technological appeal, very little progress can be made in these ASAT arms control meetings. For example, the proposals of Lt. General Daniel Graham

[9]Verbal communication received by author from Public Relations office of NORAD (North American Air Defense) in Colorado Springs.

in his new book entitled "The High Frontier" outlining a peaceful passive defense, premised on global based ballistic missiles, suggests it would seem that the United States and Russia will have to explore this ICBM defense vulnerability before progress can be made in the arms control ASAT meetings. The low cost of such a defense compels this in our stringent United States budget economy today. The same compelling economic reasons apply to the current difficulties of Russia with regards to hard currency availability. We should note that Salyut 6 and 7 with COSMOS 1267 is really a space station based missile alert. Here, we note again the nuclear balance of power in the current scene from the ABM point of view. The U.S. is also pushing a manned space station defense.

In the history of the nuclear arms race, there have been only a few periods of balance when both sides were willing to limit their appetite for new weapons. The present period may last another year, but no longer. If the opportunity is missed, the next may be a long time off, according to Carnegie Endowment. The Geostationary Orbit is thus critical in arms control and outer space, for the next decade. In regard to outer space nuclear balance of power, it must be remembered that science and Law of Outer Space have thresholds of viability for each new process. Science advances in the Geostationary Orbit very rapidly today.[10]

International space law and the "freedom of outer space" must not be impeded in its progress for the benefit of all mankind by alleged claims. It appears that the United States and Russia are substantially in agreement that international law does not recognize the alleged claims of the eight Equatorial Countries. The progress of outer space science and outer space international law for the benefit of all mankind must not be impeded by any nation, whether an LDC (Less Developed Countries) or not, seeking to advance its economic interest at the expense of all other nations, and particularly where the 1967 Outer Space Treaty stands as the guiding principle of treaty and customary international law for the peace of the entire earth. World public opinion solidly supports the 1967 Treaty.

A science-law analysis by professionals of the geostationary orbit, indicates that with the experience gained from the Gemini and the Apollo-Soyuz programs on control, guidance, and stationkeeping, there will be no need for many decades for other than I.T.U. regulations of the use of the limited geostationary orbital "slots."[11] This assumes an energy demand growth factor of 5% per annum. This also assumes that spacing can be as close as two (2) km which is technically possible now. This does not assume a physical linkage of geostationary satellites which is possible with present technology to increase the "slots" in geostationary orbit. Thus, the Bogota conflict[12] claims are postulating a problem which should not eventuate until approximately the year 2100. The Bogota claims also mix present law against future technology. There is no violation of the geostationary orbit now. The Bogota claims also represent a failure to realize that in a timely manner both law and science in outer space have in the past and will in the future advance together for the benefit of all mankind.

[10] *See*, Orbital Antenna Farms, ASTRONAUTICS AND AERONAUTICS, Sept. 1977, at 20-29.

[11] A Science-Law Analysis was made at the 3rd Princeton Space Manufacturing Conference (May 1979).

[12] *See* Bogata Declaration of Dec. 3, 1976, 6 J. SPACE L. 193 (1978).

In conclusion, there are two fundamental principles which will apply to outer space arms control in the future and this paper has attempted to outline their present status. Never forget that technological breakthroughs continue to come from the science side very rapidly in outer space and it is a tremendously dangerous destabilizing factor. We must face the fact that space is already a military arena and it is not pragmatic to say that outer space is today only for peaceful purposes. Let us hope that by future treaty implementations of the 1967 Outer Space Principles Treaty, we may ultimately be able to reach an ASAT arms control agreement for the continued peaceful purposes of outer space and for the preservation and benefit of all nations and all mankind.

LAW AND SECURITY IN OUTER SPACE: PRIVATE SECTOR INTERESTS +

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This paper is more a "think-piece" than an analytical or historical review. The role of the private sector in space was anticipated from the early days of the space age, but only in the last several years has a broad range of private sector interests begun to be pursued. In light of this increasing role, it seems timely to take up the question of private sector interests in matters of security in outer space.

Conditions in outer space pose a unique and difficult set of problems for the maintenance of peace and mutual security. The space environment itself is inhospitable to life and this requires a great deal of fragile paraphernalia merely to support living beings. Remoteness from civilization creates stresses that can outstrip the ordinary standards of civilized behavior. Moreover, in view of the needs for international cooperation, people from vastly different cultural backgrounds may be thrown together creating further stresses. And finally, as the threat of a military competition in space increases, a concomitant threat to private assets and to people in space also increases.

1. *Authority in Outer Space*

Space is inherently supra-national and its removal from sovereign control is agreed to in the 1967 Outer Space Treaty.[1] Nevertheless, there is no supra-national authority charged with maintaining security or enforcing laws or resolving disputes in the space environment. States are expected to act in conformance with the Outer Space Treaty (OST) and other relevant elements of international law and, in fact, Article VI of the OST provides that States bear international responsibility for national activities in space whether they are carried on by governmental agencies or non-governmental entities. Thus, while States have no dominion over outer space, they do bear responsibility for activities (and presumably at least civil liability for resulting damages[2]). Since police power is vested only in national governments (or their subordinate entities), the maintenance of peace and security in outer space must center on the role of these national governments. The principle of "self help" in space appears not to have been abandoned and this could support private security measures. But the conclusion is essentially that there is no recognized authority present in space charged with the responsibility and having the capability to maintain peace and security.

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+ The views expressed in this article are the author's and do not necessarily represent positions or views of the National Science Foundation, the U.S. Government, or any other organization or person.

[1]Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, Including the Moon and Other Celestial Bodies (hereinafter "Outer Space Treaty"), Jan. 27, 1967, [1967] 18 U.S.T. 2410, T.I.A.S. 6347, 610 U.N.T.S. 205 (effective Oct. 10, 1967).

[2]This has not been explored in depth sufficient to consider the matter resolved.

2. Private Sector Activities and Interests in Outer Space

While a large variety of space activities are carried out by or for governments, there are important and increasing numbers of private sector space activities. For example, communications satellites are an important part of worldwide communications and promise to play an even more dominant role in broadcasting (*e.g.*, direct broadcasting of television programs). The role of satellites in communications has become fundamental, and most communication satellites are either owned and/or operated by the private sector or have a large private sector involvement. Clearly, actions in space that affect the operation of such assets can have enormous repercussions on the economic systems of advanced nations as well as on their national security.

Private sector remote sensing activities in space are likewise on the increase. There is now the possibility that the U.S. Landsat system and the U.S. meteorological satellite system will be "privatized" in the near future.

Space manufacturing is an early phase in its development. Nevertheless, it promises to become an important element of the industrialization of space. To be sure, this kind of space activity is not likely to be prominent for decades to come; but because it could represent a great investment of funds and expertise by the private sector, its sensitivity to hostile actions is of considerable importance.

Access to space has until now been controlled exclusively by several advanced nations. In the future, we can expect to see space launch capabilities developed by several countries, some among the less developed world, and by several private sector enterprises. The proliferation of space launch capability certainly will complicate the question of security in space, first because it could make it more difficult for ground-based monitors to identify the nationality of space objects, and second because it makes the space arena more subject to conflict-laden activities.

Private sector activities in space are encouraged in order to bring the greatest benefits most efficiently to mankind. Private sector investments likely will be quite substantial and the economics and security of some nations are apt to be substantially reliant on such activities. The natural environment in space is stressful. But if space is to become an arena for resolving or responding to conflicts, whether they be conflicts between governments or between private sector entities, the resulting stresses could be enormous.

3. Governmental Agreements Concerning Security in Outer Space

Currently, agreements dealing with security and outer space primarily concern the question of how space activities can affect the security of nations. For example, the Outer Space Treaty states that activities in space shall be in the interest of maintaining international peace and security.[3] Moreover, the Treaty proscribes orbiting weapons of mass destruction,[4] which presumably are for use against masses on the Earth. The Treaty also contains provisions that deal with the uses of space *per se*.[5]

[3] Outer Space Treaty, Art. III.

[4] *Id.* Art. IV.

[5] *Id. e.g.*, Article IV. This provision declares that the Moon and other celestial bodies shall be "exclusively for peaceful purposes." Note that outer space itself is not included in this provision.

Another pact, the Accident Measures Agreement, aims at averting nuclear war between the United States and the U.S.S.R.[6] Included is an undertaking to notify the other side of any unidentified objects (including objects in outer space) that are detected by missile warning systems,[7] and of any signs of interference with missile warning systems (including those in space)[8] or of any interference with related communications.[9] The sense of the agreement might include an obligation to notify the other side if it is expected that activities in space could create the risk of nuclear war between the United States and the U.S.S.R. through misunderstanding or mistake.

The ABM Treaty[10] between the United States and the U.S.S.R. provides that the parties will not interfere with "national technical means of verification."[11] This provision raises the question of whether the term "national" is meant to include non-governmental interests, as is provided in the Outer Space Treaty.

Finally, in the Enmod (Environmental Modification) Convention,[12] signed by more than thirty countries, the parties agreed not to use environmental modification techniques,[13] on the outer space and other environments, for the destruction, damage, or injury of any other state party.[14] No mention is made of injury to private interests.

In summary, concern over protection of private sector interests is almost completely lacking in existing governmental agreements. The Outer Space Treaty does concern itself with making sure that governments are responsible for the space activities of their private sector "nationals." However, the relevant provisions in the Treaty are not without ambiguity. In particular, it is not clear what constitute non-governmental national activities in space.

4. The Impact of Private Sector Activities on Security Concerns.

Private sector assets in space are generally vulnerable to hostile action because it is quite costly to reduce their vulnerability significantly. Governmental assets, on the other hand, might be hardened or have built in redundancy sufficient to make them survivable in a reasonable range of hostile environments simply because governments

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[6]Agreement on Measures to Reduce the Risk of Outbreak of Nuclear War between the U.S. and U.S.S.R., Sept. 30, 1971, 22 U.S.T. 1590, T.I.A.S. 7186.

[7] *Id.* Art. III.

[8] *Id.* Art. III.

[9] *Id.* Art. III.

[10]Treaty Between the U.S. and U.S.S.R. on the Limitation of Anti-Ballistic Missile Systems, May 26, 1972, 23, U.S.T. 3435, T.I.A.S. 7503, effective Oct. 3, 1972.

[11] *Id.* Art. XII, para. 2.

[12]Convention on the Prohibition of Military or Any Other Hostile Use of Environmental Modification Techniques, May 18, 1977, T.I.A.S. 9614; Jan. 17, 1980.

[13] *Id.* Art. I, para. 1.

[14] *Id.* Art. II.

can afford to do so. Therefore, a private space activity can be interfered with or damaged by a less forceful and radical intervention. It is altogether possible that private activities would be the initial targets of hostile actions in space.

Many private sector space activities could be seen as provocative. For example, the direct broadcast of television to territories where the governments object could be seen to incite a response directed against the activity itself. Remote sensing, particularly by systems the products of which are of fine resolution and distributed without prior consent of the sensed state, could also be viewed as provocative.

Finally, with increasing space activity comes the threat of unintentional interference. The Liability Convention[15] provides for compensation for damages caused by space objects. For damages caused in the space environment, however, liability depends upon the demonstration of "fault." The concept of fault is difficult because its limits are not well defined juridically. In the present context, for example, it is not clear whether fault is coincident with causation or whether it excludes negligence.

5. Conclusions

It appears that private sector activities and interests in space a) are in jeopardy from hostile actions between governments, b) may provoke such actions from governments or private entities, c) are important elements in the economic health and national security of some countries, and d) are not adequately provided for in existing space law.

With increasing private investment in space, particularly from the U.S. private sector, it becomes very important for private sector concerns to be brought to bear on governmental deliberations on law and regulation in space. There exist mechanisms for bringing industrial representatives onto delegations, to backstopping teams and into the development of positions. There are also opportunities for private sector views to be heard in connection with proposed legislation or the ratification of agreements. Nevertheless, these are not now used sufficiently to ensure that private sector concerns play their appropriate role. In this matter, it would seem that both the Government and the private sector have responsibilities.

[15]Convention on International Liability for Damage Caused by Space Objects, March 29, 1972, 24 U.S.T. 2389; T.I.A.S. 7762 effective Oct. 9, 1973.